

Brittle Star-Inspired Damage Robustness: Multi-Agent Reinforcement Learning Approach

Bram Comyn, Brent Janssens, Nyah Van Wayenberge

Faculty of Science, Ghent University

1 Damage Robustness

Real-world robots might sustain damage in operation. Nature already has solutions to this problem, e.g. brittle stars:

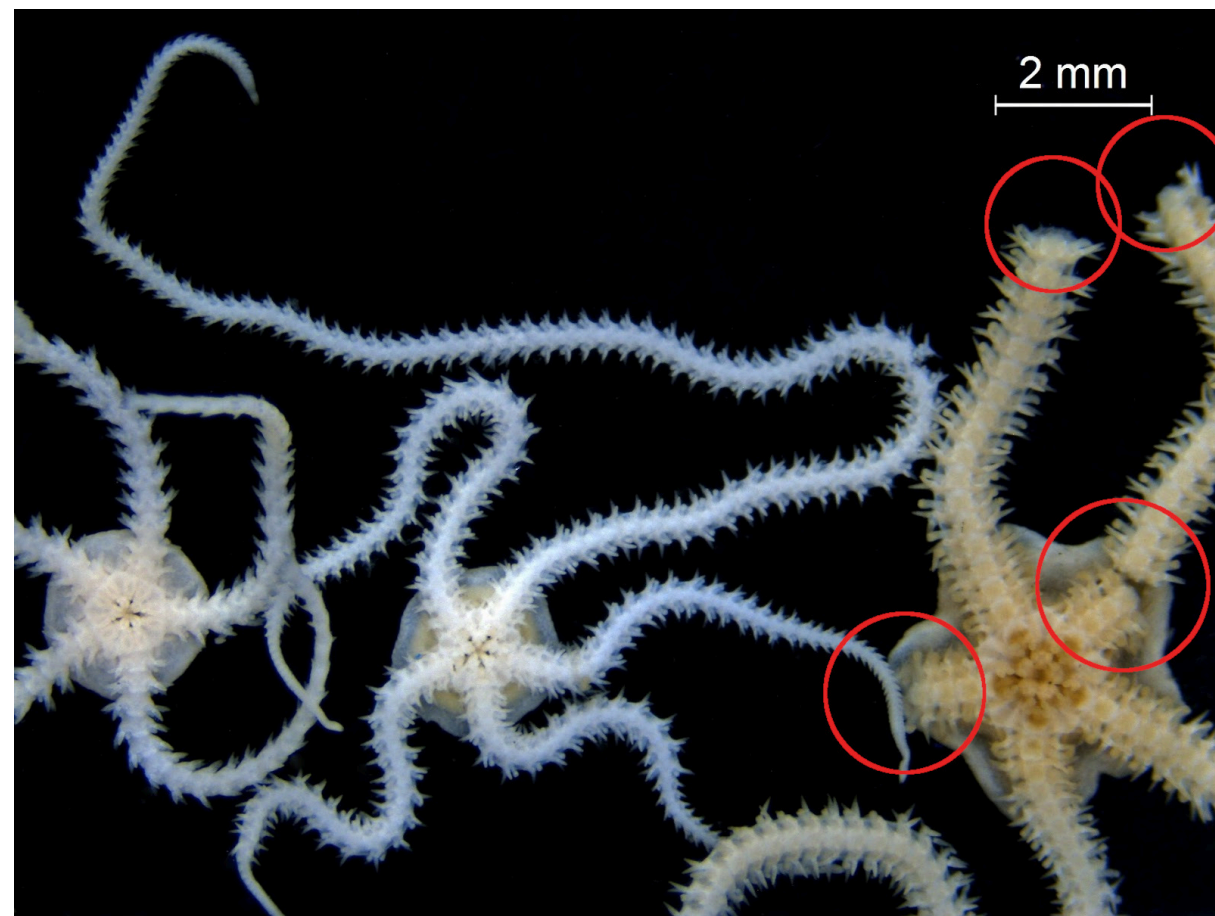


Figure 1: Two intact (left) and one damaged (right) brittle stars.



Figure 2: Brittle star-inspired robot with 5 arms in a simulated environment.

brittle star damage robustness $\xrightarrow{?}$ damage-robust robots

2 Goal

Enable damage-robust locomotion in brittle star-inspired robots using multi-agent reinforcement learning.